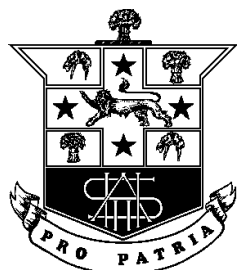


STUDENT'S NAME: \_\_\_\_\_

TEACHER'S NAME: \_\_\_\_\_



2019

TRIAL HIGHER  
SCHOOL  
CERTIFICATE  
EXAMINATION

# Mathematics Extension 1

## Assessment Task 4

---

### Examiners

- Mrs Biczó
- Ms Crancher
- Mr Potaczala
- Ms Tarannum

---

### General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- NESA-approved calculators may be used
- A Reference sheet is provided for your use
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

---

### Total marks: 70

#### Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### Section II – 60 marks (pages 5 – 10)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

**Section 1****10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section.**

Use the multiple choice answer sheet provided for Questions 1 – 10

1. Which of the following is an expression for  $\int 2 \sin^2 x \, dx$ ?

A.  $x + \frac{1}{2} \sin 2x + c$

B.  $x - \frac{1}{2} \sin 2x + c$

C.  $x + \sin 2x + c$

D.  $x - \sin 2x + c$

2. The general solution for  $\cos \theta = -\frac{1}{2}$  is given by which of the following?

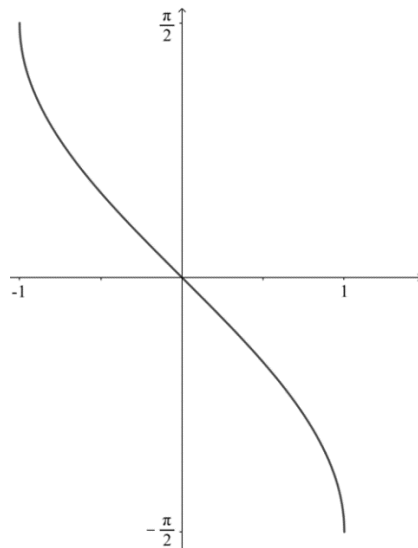
A.  $\theta = 2n\pi \pm \frac{\pi}{3}$ , where  $n$  is real

B.  $\theta = 2n\pi \pm \frac{\pi}{3}$ , where  $n$  is an integer

C.  $\theta = 2n\pi \pm \frac{2\pi}{3}$ , where  $n$  is real

D.  $\theta = 2n\pi \pm \frac{2\pi}{3}$ , where  $n$  is an integer

3. Which of the following functions best describes the graph below?



A.  $y = \sin^{-1} x$

B.  $y = -\sin^{-1} x$

C.  $y = \cos^{-1} x$

D.  $y = -\cos^{-1} x$

- A.  $y = x^3 - 2$
- B.  $y = \pm\sqrt[3]{x-2}$
- C.  $x = y^3 - 2$
- D.  $y = \sqrt[3]{x-2}$

$${}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n = (1+x)^n,$$

- A.** Differentiate both sides of the binomial expansion
- B.** Substitute  $x = 1$  into both sides of the binomial expansion
- C.** Substitute  $x = -1$  into both sides of the binomial expansion
- D.** Substitute  $x = 0$  into both sides of the binomial expansion

8. When the polynomial  $P(x) = x^3 + ax^2 + 7$  is divided by  $x + 2$ , the remainder is 11. What is the value of  $a$ ?
- A. 3                      B. -3                      C. -4                      D. 26
9. What is the solution to the inequality  $\frac{x^2 - 4}{2x} < 0$  ?
- A.  $x < -2$  and  $0 < x < 2$                       B.  $-4 < x < 0$  and  $x > 2$
- C.  $-2 < x < 0$  and  $x > 2$                       D.  $-2 < x < 0$  and  $x > 4$
10. In how many ways can 5 men and 5 women be arranged around a circular table if the women and men are to alternate?
- A. 600                      B. 2 880                      C. 14 400                      D. 86 400

**End of Section 1**

**Section 2****60 marks****Attempt Questions 11 – 14****Allow about 1 hours and 45 minutes for this section****Answer each question in a separate answer booklet.**

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11 (15 marks) Start a new answer booklet.****Marks**

(a) Prove that  $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$  **2**

(b) (i) Write  $\cos x - \sqrt{3} \sin x$  in the form  $r \cos(x + \alpha)$  **2**

(ii) Hence, find the maximum value of  $\cos x - \sqrt{3} \sin x$  and the first positive value of  $x$  for which this occurs. **2**

(c) From the top of a vertical tower, of height  $h$ , the angle of depression to a man  $M$ , standing due south of the base  $O$  of the tower, is  $42^\circ$ . From the top of the tower, another man  $G$  is observed with angle of depression  $32^\circ$ . The men are standing 500 metres apart, with  $G$  due east of  $M$ . Find the height  $h$  of the tower, to the nearest metre. **3**

(d) The curves  $f(x) = \sin x$  and  $g(x) = \cos x$  meet at  $x = \frac{\pi}{4}$ . It is given that **2**  
 $f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$  and  $g'\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$ . Calculate the acute angle between the curves at  $x = \frac{\pi}{4}$ . Give your answer in radians to two decimal places.

**Question 11 is continued on the next page**

**Question 11 continued**

(e) (i) Show that  $\tan(x+h) - \tan x = \frac{\sin h}{\cos(x+h)\cos x}$  **2**

(ii) Hence, using the first principles definition of the derivative, **2**  
 $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ , show that if  $f(x) = \tan x$ , then  $f'(x) = \sec^2 x$ .

**Question 12 is on the next page**

**Question 12 (15 marks) Start a new answer booklet.****Marks**

- (a) Consider the function  $y = \tan^{-1} \frac{1}{x}$ , where  $x \neq 0$ :
- (i) Find  $\frac{dy}{dx}$  2
- (ii) Show that  $\frac{d}{dx} \left( \tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = 0$  1
- (iii) Hence, or otherwise, sketch  $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$  for  $x > 0$  1
- (b) Using the substitution  $u = \sqrt{x}$ , find  $\int_1^9 \frac{dx}{x + \sqrt{x}}$ . 3
- (c) Show that the polynomial  $P(x) = 2x^3 - 5x^2 - 9x + 18$  has a root  $x$  such that  $1 < x < 2$ . 2
- (d) In a town, the annual growth rate of the population  $N$  is given by  $\frac{dN}{dt} = k(N - 125)$ , where  $k$  is a constant.
- (i) Show that  $N = 125 + Ae^{kt}$  is a solution to  $\frac{dN}{dt} = k(N - 125)$  2
- (ii) If the initial population was 25,650, find  $k$  given the population was 31,100 after 5 years. Round your answer to three decimal places. 2
- (e) Sketch the graph  $y = \frac{x^2}{4 - x^2}$ , clearly showing any asymptotes or intercepts with the co-ordinate axes. Your graph should be at least a quarter of a page. 2

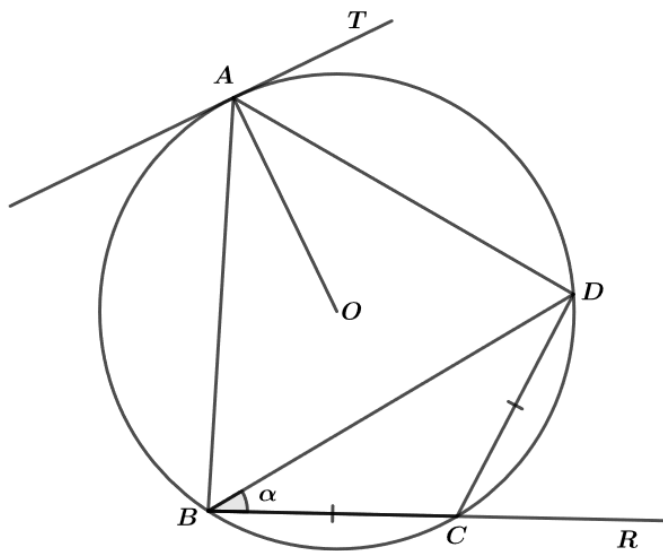
- (a) Prove that for all positive integers  $n$ ,  $9^{n+2} - 4^n$  is divisible by 5. 3
- (b) (i) By using the binomial expansion, show that 2
- $$(a+b)^n - (a-b)^n = 2 \binom{n}{1} a^{n-1} b + 2 \binom{n}{3} a^{n-3} b^3 + 2 \binom{n}{5} a^{n-5} b^5 + \dots$$
- (ii) What is the last term in the expansion when  $n$  is even? 1  
Give your answer in simplest form.
- (c) A fair six-sided die is randomly tossed  $n$  times.
- (i) Suppose  $0 \leq r \leq n$ .  
What is the probability that exactly  $r$  'sixes' appear in the uppermost position? 2
- (ii) **By using the result of part (b), or otherwise,** show that the probability that an 2  
odd number of 'sixes' appears is  $\frac{1}{2} \left\{ 1 - \left( \frac{2}{3} \right)^n \right\}$
- (d) Let  $A$  be the point  $(-3, 7)$  and let  $B$  be the point  $(1, 6)$ . Find the coordinates of the 2  
point  $P$  which divides the interval  $AB$  externally in the ratio 1 : 2.
- (e) A spherical balloon is being deflated so that the radius decreases at a constant rate of 3  
8 mm/s. Calculate the rate at which the volume is changing at the instant when the  
radius is 100 mm.

- (a) Four women and three men are available for selection in a team. A team of 4 players consists of two men and two women.

(i) How many different teams of four players can be selected? 1

(ii) Two players are husband and wife and wish to play in the same team. How many different teams can now be selected with the husband and wife on the same team? 1

(b)



*Diagram not to scale*

In the diagram above, the points  $A, B, C$  and  $D$  lie on a circle with centre  $O$ . The line  $TA$  is a tangent to the circle. The chord  $BC$  is produced to  $R$ . The interval  $AO$  bisects  $\angle BAD$  and  $BC = CD$ . Let  $\angle DBC = \alpha$ .

(i) Prove that  $\angle DCR = 2\alpha$ . 1

(ii) Show that  $\angle OAD = \alpha$ . 1

(iii) Prove that  $\angle ABC$  is a right angle. 3

**Question 14 is continued on the next page**

### Question 14 continued

- (c) Let  $\alpha, \beta$  and  $\gamma$  be the roots of  $3x^3 + 8x^2 - 1 = 0$ . **3**

What is the value of  $\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)$ ?

- (d)  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  are two points on the parabola  $x^2 = 4ay$ .

- (i) Show that the co-ordinates of  $R$ , the point of intersection of the normals at  $P$  and  $Q$  are  $(-apq(p+q), a(p^2 + pq + q^2 + 2))$ . **2**

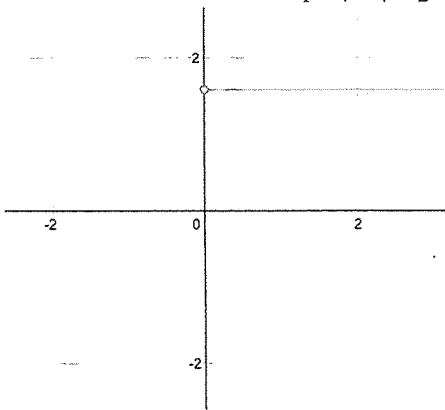
- (ii) If  $pq = -2$  find the cartesian equation of the locus of  $R$ . **3**

**End of Examination**

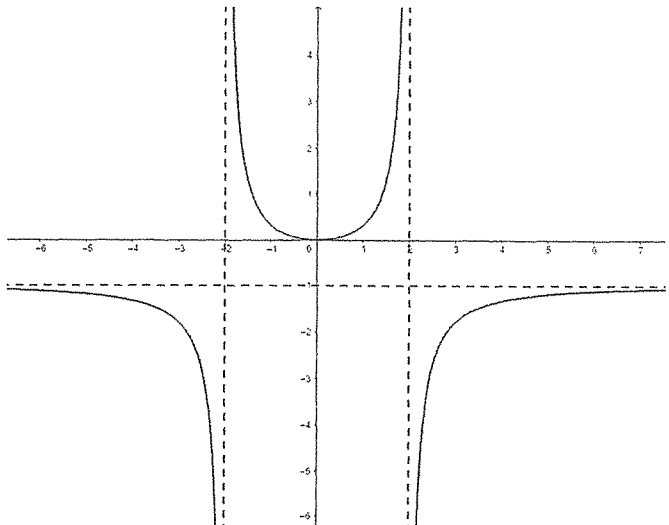
| Year 12 Trial Higher School Certificate Extension 1 Mathematics Examination 2019 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Question No. 11 Solutions and Marking Guidelines                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| Outcomes Addressed in this Question                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| PE2                                                                              | Uses multi-step deductive reasoning in a variety of contexts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                           |
| PE6                                                                              | Makes comprehensive use of mathematical language, diagrams & notation for communicating in a wide variety of situations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |
| HE1                                                                              | Appreciates interrelationships between ideas drawn from different areas of mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
| HE7                                                                              | evaluates mathematical solutions to problems and communicates them in an appropriate form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
| Outcome                                                                          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marking Guidelines                                                                                        |
| PE2                                                                              | <p>(a) Let <math>t = \tan \frac{\theta}{2}</math></p> <p>Then LHS = <math>\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}</math></p> $= \frac{1 + \frac{2t}{1+t^2} - \left(\frac{1-t^2}{1+t^2}\right)}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$ $= \frac{1+t^2+2t-(1-t^2)}{1+t^2+2t+1-t^2} \text{ (multiplying by } t^2)$ $= \frac{2t^2+2t}{2+2t}$ $= \frac{2t(t+1)}{2(1+t)}$ $= t$ $= \tan \frac{\theta}{2}$ <p><b>Note: a number of students did not state the expression on the LHS, instead starting from the second line. This can result in loss of marks in the HSC.</b></p>                                                                                                                                 | <p>2 marks : correct solution</p> <p>1 mark: significant progress toward correct solution</p>             |
| PE2                                                                              | <p>(b) (i) <math>\cos x - \sqrt{3} \sin x \equiv r \cos(x + \alpha)</math></p> $\equiv r(\cos x \cos \alpha - \sin x \sin \alpha) \text{ where}$ $r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$ $\cos x - \sqrt{3} \sin x \equiv 2 \cos x \cos \alpha - 2 \sin x \sin \alpha$ <p>Equating like coefficients, <math>1 = 2 \cos \alpha</math>, <math>-\sqrt{3} = 2 \sin \alpha</math></p> $\therefore \sin \alpha = -\frac{\sqrt{3}}{2}, \quad \cos \alpha = \frac{1}{2}$ <p>As <math>\sin</math> positive quadrants 1, 2 &amp; <math>\cos</math> positive in quadrants 1, 4 <math>\alpha</math> is in quadrant 1. <math>\therefore \alpha = \frac{\pi}{3}</math></p> $\therefore \cos x - \sqrt{3} \sin x \equiv 2 \cos\left(x + \frac{\pi}{3}\right)$ | <p>2 marks : correct solution</p> <p>1 mark : correct value for <math>r</math> or <math>\alpha</math></p> |
| PE2                                                                              | <p>(ii) Maximum value of <math>\cos x - \sqrt{3} \sin x</math> is when</p> <p><math>2 \cos\left(x + \frac{\pi}{3}\right)</math> is a maximum which occurs when</p> $\cos\left(x + \frac{\pi}{3}\right) = 1. \therefore \text{maximum value is } 2 \times 1 = 2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>2 marks : correct answers</p> <p>1 mark : one correct answer or equivalent</p>                         |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                               |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PE2, PE6 | $\cos\left(x + \frac{\pi}{3}\right) = 1 \text{ when } x + \frac{\pi}{3} = \dots, 0, 2\pi, \dots$ <p>i.e. when <math>x = \dots, -\frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \dots</math></p> <p><math>\therefore</math> maximum value is 2, and the smallest positive value of <math>x</math> for which it occurs is <math>\frac{5\pi}{3}</math>.</p> <p><b>Note: many students differentiated to find the maximum value. This is not always necessary with trig functions as <math>\sin</math> and <math>\cos</math> have maximum 1.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>3 marks : correct solution</p> <p>2 marks: substantial progress towards correct solution</p> <p>1 mark : significant progress towards correct solution</p> |
|          | <p>(c)</p> <p>Let <math>T</math> be the top of the tower.</p> <p>In <math>\Delta</math>'s <math>OTG</math> &amp; <math>OTM</math>, <math>\cot 32^\circ = \frac{OG}{h}</math> &amp; <math>\cot 42^\circ = \frac{OM}{h}</math> respectively</p> $\therefore OG = h \cot 32^\circ \text{ \& } OM = h \cot 42^\circ$ <p>As <math>G</math> is east of <math>M</math>, <math>\angle OMG = 90^\circ</math></p> $\therefore OM^2 + 500^2 = OG^2 \text{ (Pythagoras)}$ $h^2 \cot^2 42^\circ + 500^2 = h^2 \cot^2 32^\circ$ $500^2 = h^2 (\cot^2 32^\circ - \cot^2 42^\circ)$ $\therefore h^2 = \frac{500^2}{\cot^2 32^\circ - \cot^2 42^\circ}$ $\therefore h = \frac{500}{\sqrt{\tan^2 58^\circ - \tan^2 48^\circ}} \text{ using complementary angle}$ $\therefore h = 434 \text{ m results}$ <p><b>Note: Many students did not correctly position the angles of depression, or have <math>\angle OMG</math> as the right angle.</b></p> <p>(d) Using <math>\tan \theta = \left  \frac{m_1 - m_2}{1 + m_1 m_2} \right </math>,</p> $\tan \theta = \left  \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)}{1 + \frac{1}{\sqrt{2}} \times \left(-\frac{1}{\sqrt{2}}\right)} \right $ | <p>2 marks : correct solution</p> <p>1 marks : substantial progress towards correct solution</p>                                                              |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                            |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PE2         | $\tan \theta = \left  \frac{\sqrt{2}}{\frac{1}{2}} \right $ $\tan \theta = 2\sqrt{2}$ $\theta = 1.23^\circ$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                            |
| HE7,<br>HE1 | <p>(e) (i) <math>\tan(x+h) - \tan x = \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}</math></p> $= \frac{\sin(x+h)\cos x - \sin x \cos(x+h)}{\cos(x+h)\cos x}$ $= \frac{\sin((x+h)-x)}{\cos(x+h)\cos x}$ $= \frac{\sin h}{\cos(x+h)\cos x}$ <p>(ii) Given <math>f(x) = \tan x</math> and <math>f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}</math>,</p> $f'(x) = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan(x)}{h},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{\cos(x+h)\cos x \cdot h},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{\cos(x+h)\cos x} \times \frac{1}{h},$ $\therefore f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \frac{1}{\cos(x+h)\cos x},$ $= \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \lim_{h \rightarrow 0} \frac{1}{\cos(x+h)\cos x},$ $= 1 \times \frac{1}{\cos(x+0)\cos x},$ $= \frac{1}{\cos^2 x}$ $\therefore f'(x) = \sec^2 x$ <p>In show that questions, do not leave steps out. To ensure full marks, the substitution should be shown and all steps justified, such as <math>\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1</math>.</p> | <p>2 marks : correct solution<br/>1 marks : substantial progress towards correct solution</p> <p>2 marks : correct solution<br/>1 mark : substantial progress towards correct solution</p> |

|                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Year 12                                                                                                                                                                                  | Mathematics Extension 1                                                                                                                                                                                                                                                                                                     | Task 4                                                                                                                                                                                                  | Examination 2019          |
| Question No.12                                                                                                                                                                           | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                         |                           |
| <b>Outcomes Addressed in this Questions</b>                                                                                                                                              |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| H2 - constructs arguments to prove and justify results                                                                                                                                   |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| PE3 - solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations                                                 |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| HE3 - uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion or exponential growth and decay. |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| PE5 - determines derivatives which require the application of more than one rule of differentiation                                                                                      |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| HE6 - determines integrals by reduction to a standard form through a given substitution.                                                                                                 |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| H6 - uses the derivative to determine the features of the graph of a function                                                                                                            |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| HE 7 - Evaluates mathematical solutions to problems and communicates them in an appropriate form.                                                                                        |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                         |                           |
| <b>Outcomes</b>                                                                                                                                                                          | <b>Solutions</b>                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                         | <b>Marking Guidelines</b> |
| PE5                                                                                                                                                                                      | a i)<br>$y = \tan^{-1} u$ $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ $= \frac{1}{1+u^2} \times \left( -\frac{1}{x^2} \right)$ $= -\frac{1}{1+x^2}$ $u = \frac{1}{x}$ $\frac{du}{dx} = -\frac{1}{x^2}$                                                                                                             | <b>2 marks</b><br>Correct solution with appropriate working<br><br><b>1 mark</b><br>Error made<br><br>*note: if answer was inferred from part ii then 1 mark was awarded in part ii and zero for part i |                           |
| H2                                                                                                                                                                                       | ii)<br>$\frac{dy}{dx} (\tan^{-1} x + \tan^{-1} \frac{1}{x}) = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0$                                                                                                                                                                                                                        | <b>1 mark</b><br>Correct solution                                                                                                                                                                       |                           |
| H6                                                                                                                                                                                       | iii)<br>$y = \tan^{-1} x + \tan^{-1} \frac{1}{x} \text{ is a constant for } x > 0 \text{ since } \frac{dy}{dx} = 0$<br><br>When $x = 1$ then $\tan^{-1} 1 + \tan^{-1} \frac{1}{1} = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$<br> | <b>1 mark</b><br>Correct solution                                                                                                                                                                       |                           |

|           |                                                                                                                                                                                                                                                    |                                                                                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| HE6       | <p>b)</p> $x = u^2$ $\frac{dx}{du} = 2u \quad x = 1 \rightarrow u = 1$ $dx = 2u du \quad x = 9 \rightarrow u = 3$ $\int_1^3 \frac{2u du}{u^2 + u} = \int_1^3 \frac{2 du}{u + 1}$ $= [2 \ln(1 + u)]_1^3$ $= 2(\ln 4 - \ln 2)$ $= 2 \ln 2$ $= \ln 4$ | <p><b>3 marks</b><br/>Correct solution</p> <p><b>2 marks</b><br/>Error made</p> <p><b>1 mark</b><br/>Correctly changed integral limits</p>        |
| PE3       | <p>c)</p> $P(1) = 2 - 5 - 9 + 18 = 6$ $P(2) = 16 - 20 - 18 + 18 = -4$ <p>A root lies between <math>1 &lt; x &lt; 2</math> since <math>P(x)</math> is continuous and <math>P(1)</math> and <math>P(2)</math> have opposite signs.</p>               | <p><b>2 marks</b><br/>Correct solution with appropriate reasoning</p> <p><b>1 mark</b><br/>Error made or reasoning incomplete</p>                 |
| HE3<br>H2 | <p>d i)</p> $N = 125 + Ae^{kt}$ $\frac{dN}{dt} = Ake^{kt}$ $= k(125 + Ae^{kt} - 125)$ $= k(N - 125)$                                                                                                                                               | <p><b>2 marks</b><br/>Correct solution with appropriate reasoning</p> <p><b>1 mark</b><br/>Error made or reasoning incomplete / missing steps</p> |
| HE3       | <p>ii)</p> $25650 = 125 + Ae^0$ $A = 25525$ <p>so <math>N = 125 + 25525e^{kt}</math></p> $31100 = 125 + 25525e^{5k}$ $\frac{1239}{1021} = e^{5k}$ $\ln \frac{1239}{1021} = 5k \ln e$ $k = \frac{\ln \frac{1239}{1021}}{5} = 0.039$                 | <p><b>2 marks</b><br/>Correct solution</p> <p><b>1 mark</b><br/>Error made</p>                                                                    |

|     |                                                                                               |                                                                                                                                                      |
|-----|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| HE7 | <p>e)</p>  | <p><b>2 marks</b><br/>Correct solution, including smooth graph and correct behaviour of curves at asymptotes</p> <p><b>1 mark</b><br/>Error made</p> |
|-----|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

| Year 12 2019 Mathematics Extension 1 HSC Task 4-Trial Examination                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Question 13 Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |
| Outcomes Addressed in this Question                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |
| HE2 - uses inductive reasoning in the construction of proofs<br>H2 - constructs arguments to prove and justify results<br>HE3 - uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion or exponential growth and decay.<br>HE 7 - Evaluates mathematical solutions to problems and communicates them in an appropriate form. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |
| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                 | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marking Guidelines                                                                                                                            |
| HE2                                                                                                                                                                                                                                                                                                                                                                                                                     | a)<br>Step 1: Test the statement for $n = 1$<br>$9^{n+2} - 4^n = 729 - 4 = 725$ this is divisible by 5.<br><br>Step 2: Assume the statement is divisible by 5 for $n = k$<br>Thus $9^{k+2} - 4^k = 5M$ where $M$ is a positive integer.<br>Step 3: Prove the statement is divisible by 5 for $n = k + 1$ .<br><br>$9^{k+3} - 4^{k+1} = 9(9^{k+2}) - 9(4^k) + 9(4^k) - 4(4^k)$ $= 9(9^{k+2} - 4^k) + 4^k(9 - 4)$ $= 9(5M) + 5(4^k) \dots \dots \dots \text{from step 1}$ $= 5(9M + 4^k)$ $= 5N \quad \text{where } N \text{ is a positive integer.}$ Hence if the statement is divisible by 5 for $n = k$ , it is also divisible by 5 for $n = k + 1$ . It is divisible by 5 for $n = 1$ , so it is divisible by 5 for $n = 2$ . If it is true for $n = 2$ it is also true for $n = 3$ and so on. It is then true for all positive integers, $n$ . | 3 marks: complete correct proof<br><br>2 marks: substantial work that could lead to a correct proof<br><br>1 mark: correct working for step 1 |
| HE7, H2                                                                                                                                                                                                                                                                                                                                                                                                                 | b)<br>(i)<br>$(a+b)^n - (a-b)^n = \left[ a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{3}a^{n-3}b^3 + \dots + b^n \right]$ $- \left[ a^n - \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 - \binom{n}{3}a^{n-3}b^3 + \dots + (-1)^n b^n \right]$ $= 2 \left[ \binom{n}{1}a^{n-1}b + \binom{n}{3}a^{n-3}b^3 + \binom{n}{5}a^{n-5}b^5 + \dots \right]$ $\therefore (a+b)^n - (a-b)^n = 2\binom{n}{1}a^{n-1}b + 2\binom{n}{3}a^{n-3}b^3 + 2\binom{n}{5}a^{n-5}b^5 + \dots$                                                                                                                                                                                                                                                                                                                                                          | 2 marks: complete correct show<br><br>1 mark: for substantial work that could lead to a correct show                                          |
| HE7, H2                                                                                                                                                                                                                                                                                                                                                                                                                 | (ii)<br>If $n$ is even, $(-1)^n b^n$ is positive,<br>$\therefore$ last term is $2\binom{n}{n-1}ab^{n-1} = 2nab^{n-1}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 mark: correct last term if $n$ is even                                                                                                      |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HE3 | c)<br>i) $\left(\frac{5}{6} + \frac{1}{6}\right)^n$ is the binomial probability, since probability of a six $= \frac{1}{6}$ .<br><br>$\therefore$ Probability of exactly $r$ sixes, $P(r) = \binom{n}{r} \left(\frac{5}{6}\right)^{n-r} \left(\frac{1}{6}\right)^r$                                                                                                                                                                                   | 2 marks: complete correct solution<br><br>1 mark: substantial work that could lead to a correct solution                                                                                  |
| HE3 | ii) Probability of an odd number of sixes<br>$= P(1) + P(3) + P(5) + \dots$<br>$= \binom{n}{1} \left(\frac{5}{6}\right)^{n-1} \left(\frac{1}{6}\right) + \binom{n}{3} \left(\frac{5}{6}\right)^{n-3} \left(\frac{1}{6}\right)^3 + \dots$<br>$= \frac{1}{2} \left[ \left(\frac{5}{6} + \frac{1}{6}\right)^n - \left(\frac{5}{6} - \frac{1}{6}\right)^n \right], \quad \text{from b)}$<br>$= \frac{1}{2} \left[ 1 - \left(\frac{2}{3}\right)^n \right]$ | 2 marks: complete correct solution<br><br>1 mark: for substantial work that could lead to a correct solution                                                                              |
| HE7 | d)<br>$A(-3, 7) \quad B(1, 6)$<br>$-1:2$<br>$P$ has coordinates<br>$\left( \frac{(2)(-3) + (-1)(1)}{2-1}, \frac{(2)(7) + (-1)(6)}{2-1} \right)$<br>$= (-7, 8)$                                                                                                                                                                                                                                                                                        | 1 mark: for $x$ coordinate<br><br>1 mark: for $y$ coordinate                                                                                                                              |
| HE7 | e)<br>We are given $\frac{dr}{dt} = -8$ and we wish to find $\frac{dV}{dt}$<br><br>$V = \frac{4}{3}\pi r^3$<br><br>$\frac{dV}{dr} = 4\pi r^2$<br>$\therefore \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$<br>$= 4\pi r^2 (-8)$<br>$= -32\pi r^2$<br>and when $r = 100$<br>$\frac{dV}{dt} = -32\pi(100)^2$<br>$= -320\,000\pi \text{ mm}^3/\text{s}$                                                                                            | 3 marks: complete correct solution<br><br>2 marks: for finding $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = 4\pi r^2 (-8)$<br><br>1 mark: for finding $\frac{dV}{dr} = 4\pi r^2$ |

**Outcomes Addressed in this Question**

PE2 uses multi-step deductive reasoning in a variety of contexts  
PE3 solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations

| Outcome | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marking Guidelines                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PE3     | (a)<br>(i) $4C2 \times 3C2 = 18$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Award 1 mark</b> for the correct solution.                                                                                                                                                     |
|         | (ii) $3C1 \times 2C1 = 6$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Award 1 mark</b> for the correct solution.                                                                                                                                                     |
| PE2     | (b)<br>(i)<br>Given $BC = CD$<br>$\triangle BCD$ is Isosceles triangle ( two equal sides in a triangle)<br>$\angle DBC = \angle BDC = \alpha$ ( angles opposite equal sides in an Isosceles triangle are equal)<br>$\angle DCR = 2\alpha$ (exterior angle equals to the sum of interior opposite angles)                                                                                                                                                                                                                                                                   | <b>Award 1 mark</b> for the correct solution.                                                                                                                                                     |
| PE2     | (ii)<br>$\angle BAD \therefore = \angle DCR$ (exterior angle equals to the opposite interior angle in cyclic quadrilateral $BADC$ )<br>$\therefore \angle BAD = 2\alpha$<br>$\therefore AO$ bisects $\angle BAD$<br>$\angle OAD = \alpha$                                                                                                                                                                                                                                                                                                                                  | <b>Award 1 mark</b> for the correct solution.                                                                                                                                                     |
| PE2     | (iii)<br>$\angle TAO = 90^\circ$ ( angle between the tangent and radius at the point of contact )<br>$\angle TAD = 90^\circ - \alpha$ (adjacent anlg sum)<br>$\angle ABD = 90^\circ - \alpha$ ( Alternate Segment Theorem)<br>$\angle ABC = \angle ABD + \angle DBC = 90^\circ - \alpha + \alpha = 90^\circ$                                                                                                                                                                                                                                                               | <b>Award 3 marks</b> for the correct answer.<br><b>Award 2 mark</b> for substantial progress towards the correct solution.<br><b>Award 1 mark</b> for some progress towards the correct solution. |
| PE3     | (c)<br>$\alpha + \beta + \gamma = -\frac{8}{3}$<br>$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{0}{3} = 0$<br>$\alpha\beta\gamma = -\left(\frac{-1}{3}\right) = \frac{1}{3}$<br>$\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)$<br>$= \left(\alpha\beta + \frac{\alpha}{\gamma} + 1 + \frac{1}{\beta\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)$<br>$= \alpha\beta\gamma + \beta + \alpha + \frac{1}{\gamma} + \gamma + \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\alpha\beta\gamma}$ | <b>Award 3 marks</b> for the correct answer.<br><b>Award 2 mark</b> for substantial progress towards the correct solution.<br><b>Award 1 mark</b> for some progress towards the correct solution. |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | $= \alpha\beta\gamma + \frac{1}{\alpha\beta\gamma} + \alpha + \beta + \gamma + \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ $= \alpha\beta\gamma + \frac{1}{\alpha\beta\gamma} + \alpha + \beta + \gamma + \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma}$ $= \left(\frac{1}{3}\right) + \frac{1}{\left(\frac{1}{3}\right)} + \left(-\frac{8}{3}\right) + \frac{0}{\left(\frac{1}{3}\right)} = \frac{1}{3} + 3 - \frac{8}{3} = \frac{2}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                        |
| PE3 | <p>(d)<br/>(i)<br/>The equation of the normal to the parabola at <math>P</math> and <math>Q</math> is given by the equations <math>x + py = ap^3 + 2ap</math> and <math>x + qy = aq^3 + 2aq</math> respectively.<br/>Solving simultaneously<br/><math>x + py = ap^3 + 2ap \rightarrow A</math><br/><math>x + qy = aq^3 + 2aq \rightarrow B</math><br/><math>A - B</math><br/><math>(p - q)y = a(p^3 - q^3) + 2a(p - q)</math><br/><math>(p - q)y = a(p - q)(p^2 + pq + q^2) + 2a(p - q)</math><br/><math>y = a(p^2 + pq + q^2) + 2a</math><br/><math>\therefore y = a(p^2 + pq + q^2 + 2)</math><br/>Sub <math>y = a(p^2 + pq + q^2 + 2)</math> in <math>A</math><br/><math>x + pa(p^2 + pq + q^2 + 2) = ap^3 + 2ap</math><br/><math>x = ap^3 + 2ap - pa(p^2 + pq + q^2 + 2)</math><br/><math>x = ap^3 + 2ap - ap^3 - ap^2q - apq^2 - 2ap</math><br/><math>\therefore x = -apq(p + q)</math><br/><math>\therefore R(-apq(p + q), a(p^2 + pq + q^2 + 2))</math><br/>(ii)<br/><math>x = -apq(p + q)</math>, given <math>pq = -2</math><br/><math>x = 2a(p + q)</math><br/><math>(p + q) = \frac{x}{2a}</math><br/><math>y = a(p^2 + pq + q^2 + 2)</math><br/><math>y = a(p^2 - 2 + q^2 + 2)</math><br/><math>y = a(p^2 + q^2)</math><br/><math>y = a\left((p + q)^2 - 2pq\right)</math><br/><math>y = a\left(\left(\frac{x}{2a}\right)^2 - 2(-2)\right)</math><br/><math>y = \frac{x^2}{4a} + 4a</math></p> | <p><b>Award 2 marks</b> for the correct answer.</p> <p><b>Award 1 mark</b> for substantial progress towards the correct solution.</p> <p><b>Award 3 marks</b> for the correct answer.</p> <p><b>Award 2 mark</b> for substantial progress towards the correct solution.</p> <p><b>Award 1 mark</b> for some progress towards the correct solution.</p> |

Q1 B

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \quad (\text{rearranging } \cos 2x = 1 - 2\sin^2 x)$$

$$\int 2 \sin^2 x \, dx$$

$$= \int 2 \left( \frac{1}{2} (1 - \cos(2x)) \right)$$

$$= \int (1 - \cos(2x)) \, dx$$

$$= \int 1 \, dx - \int \cos(2x) \, dx$$

$$= x - \frac{1}{2} \sin 2x + c$$

Q2. D

$$\cos \theta = -\frac{1}{2}$$

$$= \dots, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3} \dots$$

$$\text{or } \theta = 2\pi n \pm \cos^{-1}\left(-\frac{1}{2}\right) = 2\pi n \pm \left(\pi - \cos^{-1}\left(\frac{1}{2}\right)\right)$$

$$= 2\pi n \pm \left(\pi - \frac{\pi}{3}\right)$$

$$\therefore 2\pi n \pm \frac{2\pi}{3}, \text{ where } n \text{ is an integer}$$

Q3. B

Q4 D

$$y = x^3 + 2$$

swap the variables ( $x$  and  $y$ )

$$x = y^3 + 2$$

Solve the equation for  $y$

$$y^3 = x - 2$$

$$y = \sqrt[3]{x-2}$$

Q5. A

Binomial expansion of  $(a + y)^4$

$$= a^4 + 4a^3y + 6a^2y^2 + 4ay^3 + y^4$$

When  $a = 2x, y = 3$

$$= 16x^4 - 96x^3 + 216x^2 - 216x + 81$$

$$\therefore -96$$

Q6 C

$$P(2F) = 5C2 (0.48)^2 (0.52)^3$$

$$= \text{combinate } 2F3M \times 2 \text{ Female} \times 3 \text{ males}$$

Q7 B

Sub in  $x = 1$  to binomial expansion given

$$\therefore \sum_{r=0}^n nCr = 2^n$$

Q8. A

$$P(-2) = -8 + 4a + 7 = 11$$

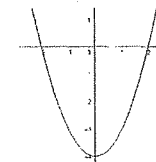
$$4a - 1 = 11$$

$$a = 3$$

Q9. A

$$\frac{x^2-4}{2x} < 0$$

$$(x+2)(x-2) < 0$$



$$0 < x < 2$$

Q10. B

$$5! \times 4! = 2880$$